An unexploitable modal agent that defects against itself

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1 Introduction

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent in Peano arithmetic, written with the provability box $\Box$ of the logic GL. FairBot, $\Box p$, cooperates exactly when it can prove that the opponent cooperates with it (p); CooperateBot always cooperates; DefectBot always defects. An agent is *unexploitable* if no modal opponent ever gets it to cooperate while the opponent defects. We consider agents built from p, from q ("the opponent cooperates with DefectBot"), $\bot$, $\neg$, $\wedge$, $\vee$ and $\Box$, with every atom under a box, and measure size as the number of nodes in the formula.

Result

The agent $W = \Box(\Box\Box\bot \vee (p \wedge (q \vee \neg\Box q)))$ is unexploitable, cooperates with FairBot and with CooperateBot, yet defects against an exact copy of itself. An exhaustive search shows that no agent of size 11 or less has this combination of properties, and W (size 12) is one of the three smallest. It is the first member of an infinite family $W_n = \Box(\Box^n\bot \vee (p \wedge (q \vee \neg\Box q)))$, $n \geq 2$, all with the same four properties, with exact laws for how W_n plays against graded FairBots $\Box^k p$, graded PrudentBots and other members of the family.

Why it matters

Every unexploitable agent in the literature we checked (FairBot, PrudentBot, CliqueBot) cooperates with its own copy. W_n shows that unexploitability, cooperation with FairBot and generosity towards CooperateBot do not force self-cooperation: the delayed "gate" $\Box^n\bot$ lets the agent tolerate FairBot's failure to prove anything about DefectBot, while two copies of W_n never reach mutual cooperation.

Proved vs checked

The properties of W_n and its laws against graded FairBots, graded PrudentBots and other W_m are proved for all indices in Lean 4 (height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [17], cited, not formalised; see also [8]). The minimality of size 12 is an exhaustive computation over all agents up to size 12 in the stated language.

Novelty statement

We believe this result is new. Our prior-art search (2026-10-17) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" (Alignment Forum).
- Mennen, "Modal Combat for games other than the prisoner's dilemma" (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the proof and check files are included so anyone can re-run them.

2 Claims

Claim K6

In L (atoms p,q boxed; F,~,&,|,[]; size=nodes), the smallest agent that cooperates with FairBot and CooperateBot, defects against itself and is unexploitable

has size 12; W=[](<>[]F|(p&(q|~[]q))) is one, unexploitable by every opponent; none of size <=11.

Status: derived.

Verified.

- `checks/K6-c046_combo12.py`: census counts to size 12; S=0,FBc=1,CBc=1 counts per size 4..12 = 2,2,4,8,19,46,120,357,900; none <=11 and exactly 3 of size 12 unexploited by all 2815 size<=9 classes; W's self/FB/CB/DB outcomes and zero exploiters by an independent height-table solver; GLS certificates for W vs W, FB, CB with flipped-candidate controls — pass

Evidence.

- reasoning: W at the tail implies p at every height >=2 (box over lower heights), so any opponent cooperates at the tail whenever W does: W is unexploitable by every modal agent, not only census agents. The 3 size-12 agents all have such a gate.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K9

For every n>=2, the agent W_n=[](<>^n F|(p&(q|~[]q))) of size n+10 is unexploitable by every modal opponent, mutually cooperates with FairBot and CooperateBot, and defects against itself (both copies defect above height n+1); W_0, W_1 fail with FairBot. Lean-proved.

Status: derived.

Verified.

- `checks/K9-c046_wn_lean.py`: embedded Lean 4 theorems w_gate, w_unexploitable, w_db, w_cb, w_self, w_self_db, w_fb compile with no sorryAx; three false variants rejected; height tables n<=12 and 3000 random opponent traces agree; size(W_n)=n+10 — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Closed GL fixed points are letter-free and so decided by height on finite linear frames (Boolos, Logic of Provability ch. 7); w_gate quantifies over arbitrary opponent traces, hence over all modal opponents.

Prior art. Possibly new: no prior art found in the owner's search.

Claim K11

For all n>=0, k>=1, j,m>=0 (height semantics of closed GL fixed points): W_n and F_k cooperate at every height iff k<n; W_n and P_j iff 2<=n<j; W_n and W_m both defect from height n+m+3. Hence all W_n (n>=2) are distinct. Lean-proved.

Status: derived.

Verified.

- `checks/K11-c046_wpairs_lean.py`: embedded Lean 4 theorems w_fk, pj_db, w_pj, w_wm compile with no sorryAx; four false variants rejected with proof errors; height tables n<=12, k,j,m<=14 (heights 0..79) agree with all theorems and K10's n=2..5 predicates — pass (Lean-proved, no sorryAx)

Evidence.

- reasoning: Closed GL fixed points are letter-free and decided by height on finite linear frames (Boolos, Logic of Provability ch. 7). F_n coops with W_m iff m>n, so F_n separates W_n from W_{n+1}.

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261017-1126`, claims K6, K9, K11, run-folder commit `662166ecff0651902cc5fcb04bf747c62d795af2`, published 2026-10-18.

3 How to verify

Each finding has its check scripts in `claims/wn-gate-family/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

4 Revisions

- 2026-10-18 — Escaped claim statements, compiled PDF, references, per-finding paper, website layout

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

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[17] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.